

SCI 4104 GEO INFORMATICS PAST EXAM PAPER

By marufugideon@gmail.com

QUESTION ONE

Assuming knowledge of ArcGIS or any other GIS packages, demonstrate your knowledge of GIS database development by answering the following two sections.

- Draw a labelled trapezoidal or parallelogram view of at least **7** thematic layers for your town of choice. **[7]**
- For the town mentioned in (a) above, complete the table below with **7** thematic layers.

Layer Name	Layer Description	Sources of Data	Tools and technologies used for documentation

[28]

- With the aid of a labelled diagram, illustrate how will you share your developed GIS database to an interested party 5000kms away. **[5]**

QUESTION TWO

- Explain **4** critical human activities that will be affected by a total satellite blackout. **[12]**
- In computing the precise GPS location, the receiver on a device uses **4** unique satellites; explain the functions of each satellite. **[8]**

QUESTION THREE

- Using appropriate examples, explain seismic hazard analysis. **[5]**
- Explain the functions of the following as used by a cartographer
 - Theodolite **[2]**
 - Digitizer **[2]**
 - GPS Tracker **[2]**
 - GIS Software **[2]**
 - Adobe Illustrator **[2]**
- Explain how remote sensing can be used in management of natural resources. **[5]**

QUESTION FOUR

- Discuss the impediments to autonomous land vehicle adoption by the general public. **[10]**
- Discuss the significance of Augmented Reality/Virtual Reality (AR/VR) technology to web-mapping. **[10]**

QUESTION FIVE

- Define the following
 - Geodesy **[1]**
 - Spatial analysis **[1]**
 - Navigation **[1]**
 - Photogrammetry **[1]**
- Explain **3** classifications of photogrammetry. **[6]**
- Explain **5** sensors that are used in remote sensing. **[10]**